Activity

Case Analysis – Avoiding Misleading Proportions

Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (in ₹ lakhs):

Month Sales

January 95

February 97

March 99

April 101

May 103

June 105

The marketing team creates a bar chart with the **Y-axis starting at 90 instead of 0**. The chart visually suggests that June sales are almost twice those of January.

Tasks:

1. Explain why the chart is misleading.
2. Identify the visualization principle that has been violated.
3. Recommend an improved visualization method.
4. Describe how this misleading chart could influence business decisions.

1. Why is the chart misleading?

The chart is misleading because the **Y-axis starts at 90 instead of 0**. This exaggerates the differences between the sales values. Although January sales are **95 lakhs** and June sales are **105 lakhs** (only about a **10.5% increase**), the chart makes June appear almost twice as large as January.

2. Visualization principle violated

- Honest representation of data
- Proportional scaling
- Avoid misleading axes (Y-axis should normally start at zero for bar charts)

3. Improved visualization

- Use a **bar chart with the Y-axis starting at 0**.
- Alternatively, use a **line chart**, since the purpose is to show monthly growth over time.

4. Influence on business decisions

The misleading chart may:

- Make management believe sales growth is much higher than reality.
- Lead to unnecessary expansion or increased production.
- Result in unrealistic sales targets.
- Misguide investors and stakeholders.

Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment in four departments.

Department Students

Computer Science 420

Electronics 280

Mechanical 200

Civil 100

The designer accidentally creates a pie chart where the slices occupy approximately **50%, 30%, 15%, and 15%** instead of the correct proportions.

Tasks:

1. Calculate the correct percentage for each department.
2. Explain why the displayed chart is misleading.
3. Suggest an alternative chart that would improve comparison accuracy.
4. Discuss how such errors could affect institutional planning.

1. Correct percentages

Total students

= 420 + 280 + 200 + 100

= 1000

Department	Students	Correct Percentage
Computer Science	**420**	**42%**
Electronics	**280**	**28%**
Mechanical	**200**	**20%**
Civil	**100**	**10%**

2. Why is the chart misleading?

The displayed pie chart shows **50%, 30%, 15%, and 15%**, which do not match the actual data. Civil Engineering is shown as **15% instead of 10%**, while Mechanical is shown as **15% instead of 20%**. This gives an incorrect impression of department sizes.

3. Better visualization

A **bar chart** is recommended because:

- Bar lengths are easier to compare.
- It provides more accurate visual comparison than pie charts.

4. Effect on institutional planning

Incorrect proportions may lead to:

- Improper allocation of faculty.
- Wrong budgeting.
- Incorrect classroom planning.
- Poor admission planning.

Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue.

Product Revenue (₹ Crores)

A 60

B 75

C 90

D 105

Instead of using equal-width bars, the designer creates bars with increasing widths, making Product D appear nearly **three times larger** than Product A.

Tasks:

1. Identify the graphical error.
2. Explain how unequal bar widths distort perception.
3. Redesign the visualization by specifying the correct design principles.
4. Discuss the possible consequences if investors rely on this chart.

1. Graphical error

The error is **using unequal bar widths** in the bar chart.

2. How unequal widths distort perception

People judge the **area** of each bar rather than only its height. Wider bars appear more important, making Product D seem much larger than Product A even though the revenue difference is only:

- Product A = ₹60 Crores
- Product D = ₹105 Crores

Actual increase

$$60105-60\times100=75\%$$

The chart exaggerates this difference.

3. Correct design principles

- Equal-width bars
- Equal spacing between bars

- Heights proportional to revenue
- Y-axis starting from zero
- Clear labels and units

4. Consequences for investors

Investors may:

- Overestimate Product D's performance.
- Invest based on misleading information.
- Misjudge company profitability.
- Lose confidence if the true figures are later revealed.

Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation.

Source Energy (GWh)

Solar 450

Source Energy (GWh)

Wind 500

Hydro 550

Biomass 520

The analyst uses a **3D column chart** with perspective effects. Due to the viewing angle, the Hydro bar appears almost **40% larger** than Wind, although the actual difference is only **10%**.

Tasks:

1. Explain why 3D charts can create misleading proportions.
2. Compare the perceived difference with the actual difference.
3. Recommend a better visualization.
4. Discuss the importance of avoiding visual distortion in sustainability reporting.

1. Why 3D charts are misleading

3D perspective changes the apparent height and volume of columns. Objects closer to the viewer appear larger, even if the numerical differences are small.

2. Actual vs perceived difference

Wind = **500 GWh**

Hydro = **550 GWh**

Actual increase

$$500550-500\times100=10\%$$

Perceived increase

≈ **40%**

Thus, the visual exaggerates the difference by about **4 times**.

3. Better visualization

Use:

- A **2D bar chart**
- A **horizontal bar chart**

These accurately represent the values without perspective distortion.

4. Importance in sustainability reporting

Accurate visualization:

- Maintains transparency.
- Builds public trust.
- Supports evidence-based policy decisions.
- Prevents incorrect investment and environmental planning.

Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five districts.

District	Cases
A	150
B	250
C	350
D	450
E	550

The analyst scales the bubble radius directly to the number of cases rather than scaling the bubble area. As a result, District E appears more than ten times larger than District A.

Tasks:

1. Explain why scaling the radius instead of the area is misleading.

2. Describe the correct method for proportional bubble sizing.

3. Suggest an alternative visualization suitable for this dataset.

4. Discuss the potential impact of this visualization on public health resource allocation.

1. Why scaling the radius is misleading

In a bubble chart, people perceive the **area** of the circles, not the radius.

If the radius is directly proportional to the number of cases:

- District A radius = 150
- District E radius = 550

Area is proportional to:

$\pi r2$

Therefore,

$(150550)2 \approx 13.4$

District E appears **over 13 times larger** than District A, while the actual cases are only:

$150550 \approx 3.67$

times greater.

2. Correct method

The **area** of each bubble should represent the data value.

Radius should be calculated as:

$r \propto \text{Value}$

This ensures the bubble area is proportional to the number of cases.

3. Alternative visualization

A **bar chart** is the best choice because:

- It is simple.
- It accurately compares case counts.
- It avoids area perception errors.

4. Impact on public health resource allocation

A misleading bubble chart may:

- Cause excessive resources to be allocated to one district.
- Underestimate the needs of other districts.
- Lead to inefficient healthcare planning.
- Affect disease control and emergency response.